

1.4.15:

Solution:

Proof. Consider S_1, S_2 subsets of a vector space V and let $\vec{x} \in \text{Span}\{S_1 \cap S_2\}$. If $S_1 \cap S_2 = \emptyset$ then $\vec{x} = \vec{0}$, which lies in every span. Otherwise, by definition

$$\vec{x} = \sum_{i=1}^n a_i s_i$$

for some $s_1, \dots, s_n \in S_1 \cap S_2$ and $a_i \in \mathbb{F}$. By definition of intersection each $s_i \in S_1$, so $\vec{x}$ is a linear combination of vectors in S_1 and hence $\vec{x} \in \text{Span}\{S_1\}$. An identical argument shows that $\vec{x} \in \text{Span}\{S_2\}$. Thus $\text{Span}\{S_1 \cap S_2\} \subseteq \text{Span}\{S_1\} \cap \text{Span}\{S_2\}$ as was to be shown. $\square$

If $S_1 = \emptyset$ then $\text{Span}\{S_1 \cap S_2\} = \text{Span}\{\emptyset\} = \{\vec{0}\} = \text{Span}\{\emptyset\} \cap \text{Span}\{S_2\}$ for any S_2 . Now to see an example in which they are not equal, take $V = \mathbb{R}^2$, $S_1 = \{(1, 0)\}$ and $S_2 = \{(2, 0)\}$ so that

$$S_1 \cap S_2 = \emptyset \Rightarrow \text{Span}\{S_1 \cap S_2\} = \{\vec{0}\}$$

but

$$\text{Span}\{S_1\} = \text{Span}\{S_2\} = \text{Span}\{S_1\} \cap \text{Span}\{S_2\} = \left\{ x \begin{pmatrix} 1 \\ 0 \end{pmatrix} : x \in \mathbb{R} \right\} \neq \{\vec{0}\}.$$

1.5.15:

Solution:

Proof. ($\Rightarrow$) First, assume S is linearly dependent. Then there exist scalars $a_1, \dots, a_n$, not all zero, such that $a_1 u_1 + \dots + a_n u_n = \vec{0}$. Let j be the largest index with $a_j \neq 0$. If $j = 1$ then $a_1 u_1 = \vec{0}$ with $a_1 \neq 0$, so $u_1 = \vec{0}$. Otherwise $j \geq 2$ and

$$u_j = -a_j^{-1} (a_1 u_1 + \dots + a_{j-1} u_{j-1}),$$

so $u_{k+1} \in \text{Span}\{u_1, \dots, u_k\}$ for $k = j - 1$.

($\Leftarrow$) Conversely, assume $u_1 = \vec{0}$ or $u_{k+1} \in \text{Span}\{u_1, \dots, u_k\}$ for some k . If $u_1 = \vec{0}$ then S is linearly dependent since $1 \cdot u_1 = \vec{0}$ is a non-trivial representation of $\vec{0}$. If $u_{k+1} \in \text{Span}\{u_1, \dots, u_k\}$ then $u_{k+1} = a_1 u_1 + \dots + a_k u_k$ for some scalars a_i , so $a_1 u_1 + \dots + a_k u_k + (-1) u_{k+1} = \vec{0}$ is a non-trivial representation of $\vec{0}$ since the coefficient of u_{k+1} is $-1 \neq 0$. Hence S is linearly dependent. $\square$

k need not be 0: the case $k = 0$ would say $u_1 \in \text{Span}\{\emptyset\} = \{\vec{0}\}$, which is exactly the case $u_1 = \vec{0}$.

2.1.16:

Solution:

Proof. In order to show that T is onto we must show that for all $f \in P(\mathbb{R})$ there exists $g \in P(\mathbb{R})$ such that $T(g) = f$. However, since T is defined by $T(f(x)) = f'(x)$ we can simply apply the fundamental theorem of calculus to take

$$g(x) = \int_0^x f(t) dt$$

so that

$$T(g(x)) = T\left(\int_0^x f(t) dt\right) = \left(\int_0^x f(t) dt\right)' = f(x).$$

Now, to show that T is not one-to-one consider $h(x) = 3x + 1$ and $k(x) = 3x + 2$ so that

$$T(h(x)) = 3 = T(k(x))$$

but $h(x) \neq k(x)$ and so T cannot be one-to-one. □

2.2.2:

Solution:

(a) We have that

$$T(e_1) = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \quad \text{and} \quad T(e_2) = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix}$$

so

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \\ 1 & 0 \end{pmatrix} = A$$

and so the matrix representation of T in the standard ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$ is the same as the matrix representation of T in the form $T = L_A$.

(b) We have that

$$T(e_1) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \text{and} \quad T(e_2) = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \quad \text{and} \quad T(e_3) = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

so

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 0 & 1 \end{pmatrix} = A$$

and so the matrix representation of T in the standard ordered bases for $\mathbb{R}^3$ and $\mathbb{R}^2$ is the same as the matrix representation of T in the form $T = L_A$.

(c) We have that

$$T(e_1) = 2 \quad \text{and} \quad T(e_2) = 1 \quad \text{and} \quad T(e_3) = -3$$

so

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 2 & 1 & -3 \end{pmatrix} = A$$

and so the matrix representation of T in the standard ordered bases for $\mathbb{R}^3$ and $\mathbb{R}$ is the same as the matrix representation of T in the form $T = L_A$.

2.2.4:

Solution: We have that

$$T(e_1) = 1 \quad \text{and} \quad T(e_2) = 1 + x^2 \quad \text{and} \quad T(e_3) = 0 \quad \text{and} \quad T(e_4) = 2x$$

so

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

2.2.5d:

Solution: We have that

$$T(e_1) = 1 \quad \text{and} \quad T(e_2) = 2 \quad \text{and} \quad T(e_3) = 4$$

$$\text{so } [T]_{\beta}^{\gamma} = \begin{pmatrix} 1 & 2 & 4 \end{pmatrix}$$

2.3.3:

Solution:

- (a) Let e_1, e_2 , and e_3 be the standard basis vectors for $P_2(\mathbb{R})$. First, in order to find $[U]_{\beta}^{\gamma}$ and $[T]_{\beta}$ we see that

$$U(e_1) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad U(e_2) = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \text{and} \quad U(e_3) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

and

$$T(e_1) = 2, \quad T(e_2) = 3 + 3x, \quad \text{and} \quad T(e_3) = 6x + 4x^2.$$

So,

$$[U]_{\beta}^{\gamma} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \quad \text{and} \quad [T]_{\beta} = \begin{pmatrix} 2 & 3 & 0 \\ 0 & 3 & 6 \\ 0 & 0 & 4 \end{pmatrix}$$

and

$$UT(e_1) = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}, \quad UT(e_2) = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}, \quad \text{and} \quad UT(e_3) = \begin{pmatrix} 6 \\ 4 \\ -6 \end{pmatrix}.$$

Thus,

$$[UT]_{\beta}^{\gamma} = \begin{pmatrix} 2 & 6 & 6 \\ 0 & 0 & 4 \\ 2 & 0 & -6 \end{pmatrix}.$$

Now, by Thm. 2.11 we have that

$$[UT]_{\beta}^{\gamma} = [U]_{\beta}^{\gamma} [T]_{\beta} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 & 0 \\ 0 & 3 & 6 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 & 6 \\ 0 & 0 & 4 \\ 2 & 0 & -6 \end{pmatrix}$$

which is exactly the matrix that we found earlier.

- (b) Clearly we have

$$[h(x)]_{\beta} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} \quad \text{and} \quad [U(h(x))]_{\gamma} = \begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix}.$$

Using Thm. 2.14 we check:

$$[U(h(x))]_{\gamma} = [U]_{\beta}^{\gamma} [h(x)]_{\beta} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix}$$

2.3.4d:

Solution: We first find $[T]_\beta^\gamma$ and $[f(x)]_\beta$:

$$[T]_\beta^\gamma = \begin{pmatrix} 1 & 2 & 4 \end{pmatrix} \quad \text{and} \quad [f(x)]_\beta = \begin{pmatrix} 6 \\ -1 \\ 2 \end{pmatrix}$$

thus

$$[T(f(x))]_\gamma = [T]_\beta^\gamma [f(x)]_\beta = \begin{pmatrix} 1 & 2 & 4 \end{pmatrix} \begin{pmatrix} 6 \\ -1 \\ 2 \end{pmatrix} = (12).$$

2.4.2:

Solution:

(c) We have that the matrix representation of T is

$$A = \begin{pmatrix} 3 & 0 & -2 \\ 0 & 1 & 0 \\ 3 & 4 & 0 \end{pmatrix} \xrightarrow{r_3 \rightarrow r_3 - r_1} \begin{pmatrix} 3 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 4 & 2 \end{pmatrix} \xrightarrow{r_3 \rightarrow r_3 - 4r_2} \begin{pmatrix} 3 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

and so clearly $\text{rank}(A) = 3 = \dim \mathbb{R}^3$ and thus T is invertible by Thm. 2.5.

(d) As shown in ex(2.1.16), T is not one-to-one and thus cannot be invertible.

2.4.7:

Solution:

(a) *Proof.* Suppose for the sake of contradiction that A is invertible. Then,

$$\begin{aligned} A^2 = 0 &\Rightarrow AA = 0 \Rightarrow A^{-1}(AA) = A^{-1}0 \\ &\Rightarrow (A^{-1}A)A = 0 \\ &\Rightarrow IA = 0 \\ &\Rightarrow (IA)A^{-1} = 0A^{-1} \\ &\Rightarrow I(AA^{-1}) = 0 \\ &\Rightarrow II = 0 \\ &\Rightarrow I = 0 \end{aligned}$$

which is a contradiction and thus A is not invertible. □

(b) A cannot be invertible, for otherwise

$$AB = 0 \Rightarrow A^{-1}(AB) = A^{-1}0 \Rightarrow (A^{-1}A)B = 0 \Rightarrow IB = 0 \Rightarrow B = 0$$

which is a contradiction as B is specifically nonzero and so A cannot be invertible.

⁰LaTeX code for this document can be found on github [here](#)